Quantifying the Environmental Cost of AI in Contact Center Operations

An Energy Analysis of Agentic Email, AI-Assisted Voice, and the Human Infrastructure They Augment or Replace

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Abstract

As enterprises accelerate AI adoption across customer contact operations, a growing but underexamined question is the environmental cost of this shift. This paper provides a first-order energy analysis of two common AI-powered contact center workflows—agentic email processing and AI-assisted live voice calls—and compares their energy footprints to the human infrastructure they augment or replace. Drawing on the most recent empirical data from OpenAI, Google, Epoch AI, and academic benchmarking studies (2024–2025), we estimate per-transaction energy consumption for each workflow, then scale these estimates to representative enterprise volumes. We find that while individual AI transactions consume measurably more energy than a traditional search query, the incremental environmental cost is modest in absolute terms, and in scenarios where AI fully resolves interactions, it can represent a net reduction in total energy expenditure compared to human-only operations when facility, commute, and work-from-home energy costs are considered.

1. Introduction: The Question Nobody Has an Answer To

At a recent customer operations town hall, an employee asked a deceptively simple question: *"What are we doing to mitigate the environmental cost of our increasing AI usage?"* The leadership team—including the author—had no real answer. This paper is an attempt to build one.

The question is timely. The International Energy Agency projects that global data center electricity consumption will roughly double between 2024 and 2028, driven substantially by AI workloads. Tech companies are investing hundreds of billions in AI infrastructure. Yet for most enterprises deploying AI into operational workflows, the environmental dimension remains unquantified—a blind spot in otherwise rigorous deployment planning.

This paper addresses that gap by analyzing two specific, real-world contact center AI workflows: agentic email processing (where an AI system reads inbound email, determines intent, and drafts a response) and AI-assisted live voice calls (where AI transcribes, classifies intent, scores sentiment, and generates a summary in real time). These represent the core of what many enterprises—including medical device companies, financial services firms, and technology providers—are deploying today.

2. Methodology and Data Sources

Estimating the energy consumption of AI inference is an evolving science. No two queries are identical, and the energy cost varies with model size, hardware, data center efficiency (PUE), and the complexity of the task. This paper draws on the best available empirical data as of early 2026:

- **OpenAI disclosure (2025):** Sam Altman reported an average ChatGPT query consumes approximately 0.34 Wh of electricity, a figure broadly corroborated by independent researchers.
- **Google environmental report (August 2025):** Google published that the median Gemini Apps text prompt consumes 0.24 Wh of energy and emits 0.03 grams of CO_2 equivalent.

- **Epoch AI analysis (February 2025):** Estimated a typical ChatGPT query at approximately 0.3 Wh, noting that the widely cited figure of 3 Wh is likely an overestimate by an order of magnitude.
- **Jegham et al. academic benchmark (May 2025):** Measured GPT-4o at approximately 0.42 Wh for a short query, with reasoning models like o3 consuming over 33 Wh per complex prompt—a 70x range depending on model and task.
- **MIT Technology Review investigation (May 2025):** Estimated approximately 1 Wh for an average Llama text query and noted that agentic and reasoning workloads will drive substantially higher per-transaction energy as AI moves from simple Q&A toward multi-step autonomous workflows.

For the contact center workflows analyzed here, we use conservative midpoint estimates that account for the fact that enterprise AI tasks are more complex than a typical consumer chatbot query. Agentic email involves multiple inference passes (comprehension, classification, generation, quality check), and voice-assisted calls involve continuous ASR processing plus multiple downstream inference tasks.

3. Scenario 1: Agentic Email Processing

3.1 Workflow Description

In an agentic email workflow, an AI system receives an inbound customer email, determines the customer's intent, retrieves relevant context (e.g., order status, account information), and drafts a response for agent review or automatic dispatch. A typical transaction involves 3,800–4,200 tokens of combined input and output, with multiple inference steps including comprehension, intent classification, context retrieval prompting, and response generation.

3.2 Energy Estimate

Given the multi-step nature of agentic email (typically 2–4 chained inference calls), and calibrating against the published benchmarks above, we estimate a per-transaction energy cost of approximately 1.5–4.0 Wh (0.0015–0.004 kWh). This is roughly 4–10x a simple consumer chatbot query, reflecting the added complexity of classification, retrieval augmentation, and generation steps. We use a midpoint of 2.5 Wh (0.0025 kWh) for scaling calculations.

3.3 Scaled Impact

Daily Volume	Daily kWh	Annual MWh	Annual CO_2 (kg)	Homes Equiv.
5,000 emails	12.5	4.6	1,800	~0.4
20,000 emails	50	18.3	7,300	~1.7
50,000 emails	125	45.6	18,250	~4.2

Table 1. Agentic email energy scaling. CO_2 calculated at US grid average ~0.4 kg/kWh. Household equivalent based on US average 10,500 kWh/year.

4. Scenario 2: AI-Assisted Voice Call

4.1 Workflow Description

In an AI-assisted voice workflow (representative of platforms like Observe.ai), the AI system performs five distinct compute tasks during a single call with an average handle time (AHT) of 6 minutes: (1) real-time speech-to-text transcription running continuously for the call duration, (2) intent classification from the transcript, (3) triggering a database query based on the classified intent, (4) sentiment scoring of the interaction, and (5) generating a call summary written to Salesforce.

4.2 Per-Component Energy Breakdown

Component	Est. Energy (Wh)	Notes
ASR (6 min continuous)	1.5–3.0	Most sustained compute; model runs full duration
Intent classification	0.5–1.5	Single LLM pass over ~1,500–2,500 token transcript
Database query logic	~0.1–0.3	Lightweight decision; actual DB call is traditional compute
Sentiment scoring	0.3–1.0	Often runs as secondary classification during intent pass
Summary generation to SF	1.0–2.5	Generative output: 300–600 tokens from full transcript
Total per call	**3.4–8.3**	Midpoint: ~5.5 Wh (0.0055 kWh)

Table 2. Per-component energy estimate for a 6-minute AI-assisted voice call. ASR estimates calibrated against Whisper-class models and cloud ASR benchmarks.

4.3 Scaled Impact

Daily Volume	Daily kWh	Annual MWh	Annual CO_2 (kg)	Homes Equiv.
5,000 calls/day	27.5	10.0	4,000	~1.0
20,000 calls/day	110	40.2	16,060	~3.8
50,000 calls/day	275	100.4	40,150	~9.6

Table 3. AI-assisted voice call energy scaling at 0.0055 kWh midpoint per call.

5. The Other Side of the Ledger: Human Infrastructure Costs

Any honest environmental analysis of AI must compare the incremental energy cost of AI inference against the energy cost of the human infrastructure it augments or replaces. This is where the conversation becomes strategically interesting.

5.1 The In-Office Agent

A contact center agent working in a physical office building consumes energy across multiple dimensions: their workstation (monitors, desktop/thin client, headset), shared facility overhead (lighting, HVAC, networking equipment), and the embodied energy of the building itself. Reasonable estimates for an agent's share of facility energy run approximately 1.0–1.5 kW averaged across a full shift, yielding roughly 0.10–0.15 kWh for a 6-minute call interaction.

5.2 The Work-From-Home Agent

The most favorable human baseline for energy comparison—eliminating commute entirely—is a work-from-home agent. Even in this scenario, the agent consumes incremental household energy: a home workstation (~60–120W), additional lighting, and the marginal HVAC load of occupying a room during working hours. In a climate like the DFW metro area, summer air conditioning alone can add 0.5–1.0 kWh per work hour. A conservative estimate for the incremental home office energy footprint is 0.8–1.5 kWh per hour, or approximately 0.06–0.12 kWh per 6-minute call interaction. This means that even under the most energy-efficient human work arrangement, a single agent's home office footprint per call is 11–22 times the energy cost of the AI layer processing that same call.

5.3 The Commute (Supplementary Context)

For agents who do commute, the energy overhead is substantial. An average round trip of 40 miles in a gasoline vehicle averaging 27 mpg consumes approximately 1.48 gallons of fuel. This is important: the full chemical energy of that fuel—49.9 kWh per day at 33.7 kWh per gallon—represents a real environmental cost. The atmosphere absorbs the complete combustion output regardless of how much useful work reaches the wheels. Only ~25% of gasoline's chemical energy converts to motion; the remaining ~75% is released as waste heat, but the CO_2 emissions and thermal load are products of burning the full gallon. For an EV commuter at 0.32 kWh/mile, the round trip consumes approximately 12.8 kWh of electrical energy. In either case, the commute energy for a single agent on a single day exceeds the AI layer's energy cost for processing that agent's entire daily call volume by two to three orders of magnitude. However, commute is included here as supplementary context rather than the primary comparison, as work arrangements vary across organizations.

5.4 Comparative Summary

The following table compares per-call energy costs across scenarios. The work-from-home baseline represents the most energy-favorable human arrangement and serves as the primary comparison. Commute scenarios use full chemical energy (not useful work at the wheels) because the environment absorbs the total combustion cost.

Scenario	kWh per 6-min Call	Multiple of AI Cost
AI inference only (voice-assisted)	~0.0055	1.0x (baseline)
WFH agent (no commute)	~0.06–0.12	11–22x
In-office agent (no commute)	~0.10–0.15	18–27x
+ Gas commute, 40 mi RT (full chemical energy)	~0.72–0.77	131–140x
+ EV commute, 40 mi RT (wall-plug energy)	~0.26–0.31	47–56x

Table 4. Energy cost comparison per 6-minute call interaction. WFH is the primary comparison baseline. Gas commute uses full 49.9 kWh/day chemical energy (33.7 kWh/gal × 1.48 gal) allocated across 80 calls. EV commute uses 12.8 kWh wall-plug energy allocated across 80 calls.

The key finding: Even under the most energy-favorable human work arrangement—a work-from-home agent with no commute—the incremental energy cost of the home office (equipment, lighting, climate control) exceeds the AI layer's energy cost per call by 11–22x. The AI inference layer adds approximately 3–5% incremental energy on top of a WFH agent's footprint for an assisted call. In scenarios where AI fully resolves interactions without human involvement, the AI-only energy cost is 11–22x lower than even the most efficient human alternative.

6. Strategic Implications

6.1 The Reframe: AI-Resolved Is a Net Environmental Win

The data supports a counterintuitive conclusion: well-implemented AI automation in contact centers is likely to *reduce* an organization's total energy footprint, not increase it, in proportion to the share of interactions that AI fully resolves without human involvement. Every call or email that AI handles end-to-end eliminates the personal energy overhead of a human agent for that interaction—including their home office equipment, climate control, and lighting—regardless of whether they work on-site or remotely.

6.2 The Cumulative Concern Remains Valid

This finding should not be used to dismiss environmental concerns about AI. The concern is real at the macro level: global data center energy consumption is growing rapidly, driven by AI inference at scale. But at the enterprise level, the environmental impact of deploying AI into contact center workflows is not the catastrophic additive cost that some narratives suggest. The honest answer is more nuanced: individual enterprise AI deployments are environmentally modest, but the collective global scale of AI adoption demands industry-wide attention to infrastructure efficiency and renewable energy sourcing.

6.3 Actionable Levers

Organizations deploying AI can take concrete steps to minimize environmental impact:

1. **Cloud region selection:** Route inference workloads to data center regions powered by high percentages of renewable energy. Azure, AWS, and GCP all publish carbon intensity by region.

2. **Vendor accountability:** When evaluating AI vendors, request disclosure of hosting regions, PUE metrics, and renewable energy commitments.
3. **Efficiency as sustainability:** AI systems that handle interactions accurately on the first pass consume less total compute than implementations requiring retries, escalations, and human rework. Quality and environmental responsibility are aligned.
4. **Model right-sizing:** Not every task requires the largest available model. Intent classification and sentiment scoring can often be handled by smaller, more efficient models, reserving larger models for complex generative tasks.
5. **Measurement and disclosure:** Even rough estimates demonstrate awareness and governance maturity. Organizations that can quantify their AI energy footprint—even imperfectly—are better positioned for ESG reporting, regulatory compliance (particularly under emerging EU Scope 3 frameworks), and stakeholder communication.

7. Conclusion

The environmental cost of AI in contact center operations is real, measurable, and—at current scales—surprisingly modest. A single AI-assisted voice call consumes roughly the energy equivalent of running a standard LED light bulb for about half an hour, or approximately 20 seconds of microwave operation. Scaled to tens of thousands of daily interactions, the annual energy footprint remains in the range of a few US households—orders of magnitude smaller than the human infrastructure it operates alongside, even in a fully remote work arrangement.

The more important insight is structural: AI-assisted operations are marginally additive to the existing human energy footprint, while AI-resolved operations are substantially reductive. For enterprises genuinely concerned about environmental impact, the most impactful action is not to slow AI adoption, but to ensure that AI is deployed thoughtfully—with efficient models, renewable-powered infrastructure, and governance that treats environmental cost as a first-class deployment metric alongside accuracy, latency, and cost-per-interaction.

The employee who asked the question at the town hall deserves a real answer. This paper is a start.

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